LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 2 Test (Set B): Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following gives carbon dioxide with dilute hydrochloric acid? **[1]**

(a) Zinc (b) Sodium hydroxide
(c) Sodium hydrogencarbonate (d) Sodium chloride

Q2. Which of these has the highest pH? **[1]**

(a) Lemon juice (b) Milk of magnesia
(c) Pure water (d) Vinegar

Q3. Which indicator turns pink in a basic solution? **[1]**

(a) Methyl orange (b) Blue litmus
(c) Turmeric (d) Phenolphthalein

Q4. The chemical formula of Plaster of Paris is: **[1]**

(a) $CaSO_4 \cdot 2H_2O$ (b) $CaSO_4 \cdot \frac{1}{2}H_2O$
(c) $CaSO_4$ (d) $CaCO_3$

Q5. The stinging hairs of nettle leaves inject: **[1]**

(a) citric acid (b) oxalic acid
(c) lactic acid (d) methanoic acid

Q6. **Assertion (A):** Tooth decay starts when the pH in the mouth falls below 5.5.
Reason (R): Below pH 5.5, tooth enamel starts to get corroded. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What is a universal indicator? Why is it more useful than litmus? **[2]**

OR

Hydrochloric acid and acetic acid of the same concentration are taken. Why is HCl a stronger acid?

Q8. Why should Plaster of Paris be stored in a moisture-proof container? **[2]**

SECTION C

6 marks

Q9. A solution has a pH of 3. **[3]**

(i) Is the solution acidic, basic or neutral? **[1]**

(ii) What happens to its pH when it is diluted with water? **[1]**

(iii) What happens to the pH if a base is added slowly until the solution is neutralised? **[1]**

Q10. How is baking powder different from baking soda? Why does baking powder make a cake soft and spongy? **[3]**

OR

Write balanced equations, and name the salt formed, for the reactions of: (i) hydrochloric acid with sodium hydroxide, (ii) sulphuric acid with potassium hydroxide, (iii) nitric acid with calcium hydroxide.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) What happens when carbon dioxide is passed through lime water, and when excess carbon dioxide is passed? Write the equations. **[2]**

(ii) Give three examples of neutralisation in everyday life and explain each briefly. **[3]**

OR

(a) What is water of crystallisation? Describe what happens when blue copper sulphate crystals are heated, and when water is then added. [3]
(b) Name the salt used (i) to remove permanent hardness of water, (ii) to disinfect drinking water. [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Plants grow well only when the soil pH is in a suitable range. A farmer found that the soil in his field had a pH of 5. In his area, rain water sometimes has a pH below 5.6, which is called acid rain. When acid rain flows into rivers, it lowers the pH of the river water. **[4]**

(i) Is the farmer's soil acidic or basic? **[1]**

(ii) What can he add to the soil to correct its pH? **[1]**

(iii) Why is it a problem when acid rain lowers the pH of river water? Why is it important to keep soil at the right pH? **[2]**

~ All the best! ~